Physical Chemistry Cume

October 10, 2008

Rotational absorption lines of $^{1}H^{35}Cl$ were found in the microwave spectrum at (in cm^{-1}) 83.32, 104.13, 124.73, 145.37, 165.89, 186.23, 206.60.

a)Calculate the energies (in cm^{-1}) of the first five rotational levels of $^{1}H^{35}Cl$ and their degeneracies.

b)Calculate the bond distance of $^{1}H^{35}Cl$.

c)What is the value of the rotational quantum number, J, in the most highly populated rotational level of $^{1}H^{35}Cl$ at room temperature (T=300K)?

Apply the rigid rotor approximation. That is, with the usual symbols, $E_J=BJ(J+1)$; $B=h/(8\pi^2Ic)$; h=6.626E-34Js; c=3E8 m/s; masses are H=1 u; Cl =35 u.